BORCEA–BRÄNDÉN AIM PROBLEMS

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Abstract. This note records four statements refuted in the *Count ex Machina* archive: each is quoted as posed and answered by an explicit witness, and the archive carries a certificate that recomputes the refutation by exact arithmetic.

Summary: Borcea–Brändén Problem 35: common-matrix mixed-determinant representation (2009) *Certificate:* Stable degree-five polynomial plus size-five immanant POT obstruction.

Summary: Borcea–Brändén Problem 38: Stable Permanent-On-Top (2009) *Certificate:* Exact Schur expansion; four violating partitions.

Summary: Borcea–Brändén Problem 36: stable Schur positivity (2009) *Certificate:* Positive-definite 5×5 det-pencil; exact Schur expansion with $[s_{(1^5)}]q = -2972$.

Summary: Borcea–Brändén Problem 37: stable Schur inequality (lower endpoint) (2009) *Certificate:* Schur-positive member of the same family; $a_{(1^5)} = 69 < 125 = f^{(1^5)} a_{(5)}$.

1. Problems 35 and 38: mixed-determinant representation and Permanent-On-Top

Posed by Borcea and Brändén (2009); counterexample found by GPT-5 (Pro) (2026-01-10); formalized by S. Sra; audited by S. Sra.

A polynomial $f \in \mathbb{R}[z_1, \ldots, z_n]$ is *real stable* if $f(z_1, \ldots, z_n) \neq 0$ whenever every z_i has strictly positive imaginary part. A homogeneous symmetric polynomial of degree d expands uniquely in the basis of Schur polynomials as $p = \sum_{\lambda \vdash d} a_\lambda s_\lambda$; we say p is *Schur-positive* if every $a_\lambda \geq 0$. For a partition $\lambda \vdash d$ we write f^λ for the number of standard Young tableaux of shape λ, λ' for the conjugate partition, and, for a $d \times d$ matrix $A = (a_{ij})$,

$$\mathrm{Imm}_\lambda(A) := \sum_{\sigma \in S_d} \chi^\lambda(\sigma) \prod_{i=1}^{d} a_{i\sigma(i)},$$

for the immanant induced by the irreducible character χ^λ of S_d. The two extreme cases are $\mathrm{Imm}_{(1^d)} = \det$ and $\mathrm{Imm}_{(d)} = \mathrm{per}$. Next, for $n \times n$ matrices $A_1, \ldots, A_m$ the *mixed-determinant* is

$$\eta(A_1, \ldots, A_m) := \sum_{(S_1, \ldots, S_m)} \det(A_1[S_1]) \cdots \det(A_m[S_m]),$$

where the sum runs over all ordered partitions of $\{1, \ldots, n\}$ into disjoint sets $S_1, \ldots, S_m$, and $A[S]$ denotes the principal submatrix of A with rows and columns in S. When $A \succeq 0$ the polynomial $\eta(z_1 A, \ldots, z_n A)$ is symmetric, homogeneous, real stable, and has the (Schur-positive) representation

$$\eta(z_1 A, \ldots, z_n A) = \sum_{\lambda \vdash n} \mathrm{Imm}_{\lambda'}(A) s_\lambda(z_1, \ldots, z_n).$$

The two problems below ask how far those properties go the other way.

Date: 20 August 2026.

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From the *Count ex Machina* archive of counterexamples, https://github.com/suvrit/count-ex-machina, whose companion paper is arXiv:2608.29595.

Problem 1.1 (As posed: Borcea–Brändén, AIM problem list (2009), Problem 35)**.** Characterize all symmetric, real stable and homogeneous polynomial of degree d in n variables. Is it in fact so that if p is such a polynomial, then there is a $d \times d$ positive semidefinite $d \times d$ matrix A such that

$$p(z_1, \ldots, z_n) = \eta(z_1 A, \ldots, z_n A)?$$

This is not obvious even for $n = 2$.

Problem 1.2 (As posed: Borcea–Brändén, AIM problem list (2009), Problem 38)**.** Lieb's Permanent-On-Top (POT) Conjecture for the symmetric group on d elements [J,L] asserts that if A is a $d \times d$ positive semi-definite matrix then $\mathrm{Imm}_\lambda(A) \le f^\lambda \operatorname{per}(A)$ for all $\lambda \vdash d$. An affirmative answer to the following problem would in particular imply the validity of the POT Conjecture.
Problem 38. Let p be as in Problem 36. Is $a_\lambda \le f^\lambda a_{(1^d)}$ for all $\lambda \vdash d$?

We answers to the above problems are negative. Let $S = x_1 + \cdots + x_5$ and define the polynomial

$$p(x_1, \ldots, x_5) = \prod_{i=1}^{5} \Big(x_i + 5 \sum\nolimits_{j \neq i} x_j \Big) = \prod_{i=1}^{5} (5S - 4x_i). \tag{1.1}$$

Every factor has strictly positive coefficients, hence has positive imaginary part whenever all variables do; therefore p is real stable. Moreover, p is symmetric, homogeneous of degree five in five variables, and has strictly positive monomial coefficients.

Theorem 1.3 ([**Disproof**]: Stable Permanent-On-Top, AIM Problem 38)**.** *The polynomial* (1.1) *satisfies all hypotheses of Problem 1.2 but violates* $a_\lambda \le f^\lambda a_{(1^5)}$.

Proof. A short explicit calculation shows that p has the Schur expansion

$$p = 625 s_{(5)} + 4500 s_{(4,1)} + 7125 s_{(3,2)} + 8150 s_{(3,1,1)} + 7525 s_{(2,2,1)} + 6580 s_{(2,1,1,1)} + 1281 s_{(1^5)}.$$

For $\lambda = (3,2)$, the hook-length formula gives $f^{(3,2)} = 5$, whereas

$$7125 > 5 \cdot 1281 = 6405.$$

The same upper endpoint fails for $(3,1,1)$, $(2,2,1)$, and $(2,1,1,1)$. The accompanying source package reconstructs the monomial expansion and inverts the Kostka matrix over $\mathbb{Z}$. □

The polynomial also has a diagonal positive-definite mixed-determinant representation. For $j = 1, \ldots, 5$, let D_j denote $5I$ except for the j-th entry being just 1. Then the multivariate Johnson polynomial satisfies

$$\eta(x_1 D_1, \ldots, x_5 D_5) = p(x_1, \ldots, x_5).$$

Theorem 1.4 ([**Disproof**]: common-matrix representation, AIM Problem 35)**.** *There is no* 5×5 *Hermitian positive semidefinite matrix* A *such that* $p(x_1, \ldots, x_5) = \eta(x_1 A, \ldots, x_5 A)$.

Proof. Since $\eta(x_1 A, \ldots, x_5 A)$ is Schur positive with immanants as coefficients, the representation $p(x) = \eta(x)$ would identify the Schur coefficients (up to normalization). Since Lieb's permanental dominance inequality is known to hold in size five (indeed in size ≤ 13 (Wanless, 2022) it is known), it would force $a_\lambda \le f^\lambda a_{(1^5)}$ for every $\lambda \vdash 5$. But this contradictions Theorem 1.3, hence there cannot be any PSD A for which $p(x) = \eta(x_1 A, \ldots, x_5 A)$. □

Remark 1.5 (Scope). The example is Schur-positive, and its lower endpoint inequalities hold, so it refutes neither Problem 2.1 nor Problem 3.1; those require the positive-definite pencils of Theorems 2.2 and 3.2 below. The full independent audit is included as `counterexamples/aim-problems/artifacts/`.

2. Problem 36: stable Schur positivity

Theorem 2.2: Posed by Borcea and Brändén (2009); counterexample found by GPT-5.6 (Pro) (2026-07-30); formalized by S. Sra; audited by S. Sra.

We keep the notation of the previous section. Problem 2.1 asks whether stability forces a Schur-positive representation.

Problem 2.1 (As posed: Borcea–Brändén, AIM problem list (2009), Problem 36)**.** Let p be a symmetric, real stable and homogeneous polynomial of degree d in n variables with $d \le n$ and suppose that p has at least one positive (and then all non-negative) coefficients in the monomial basis. Is p Schur-positive? That is, when $p(z_1,\dots,z_n)$ is expanded in the Schur-basis

$$p(z_1,\dots,z_n) = \sum_{\lambda \vdash d} a_\lambda s_\lambda(z_1,\dots,z_n),$$

is $a_\lambda \ge 0$ for all $\lambda \vdash d$?

The answer to Problem 2.1 is no already at $d = n = 5$. Let

$$J_1 = \begin{pmatrix} 38 & 19 & 19 & 2 & 36 \\ 19 & 38 & 2 & 19 & 36 \\ 19 & 2 & 38 & 19 & 36 \\ 2 & 19 & 19 & 38 & 36 \\ 36 & 36 & 36 & 36 & 72 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 38 & 19 & 19 & 17 & 21 \\ 19 & 38 & 17 & 19 & 21 \\ 19 & 17 & 38 & 34 & 36 \\ 17 & 19 & 34 & 38 & 36 \\ 21 & 21 & 36 & 36 & 42 \end{pmatrix}, \quad J_3 = \begin{pmatrix} 38 & 34 & 19 & 17 & 36 \\ 34 & 38 & 17 & 19 & 36 \\ 19 & 17 & 38 & 19 & 21 \\ 17 & 19 & 19 & 38 & 21 \\ 36 & 36 & 21 & 21 & 42 \end{pmatrix},$$

$$J_4 = \begin{pmatrix} 38 & 4 & 19 & 2 & 21 \\ 4 & 8 & 2 & 4 & 6 \\ 19 & 2 & 38 & 4 & 21 \\ 2 & 4 & 4 & 8 & 6 \\ 21 & 6 & 21 & 6 & 42 \end{pmatrix}, \quad J_5 = \begin{pmatrix} 8 & 4 & 4 & 2 & 6 \\ 4 & 38 & 2 & 19 & 21 \\ 4 & 2 & 8 & 4 & 6 \\ 2 & 19 & 4 & 38 & 21 \\ 6 & 21 & 6 & 21 & 42 \end{pmatrix},$$

and define the determinantal polynomial

$$q(x_1,\dots,x_5) = \frac{1}{648} \det\Big(\sum_{i=1}^5 x_i J_i\Big). \tag{2.1}$$

Theorem 2.2 (**[Disproof]**: stable Schur positivity, AIM Problem 36)**.** *The polynomial* (2.1) *is symmetric, homogeneous of degree five in five variables, real stable, and all* 126 *of its ordinary monomial coefficients are strictly positive, yet it is* **not** *Schur positive, since*

$$[s_{(1^5)}]\, q = -2972 < 0.$$

Proof. The leading principal minors of J_i are

	Δ_1	Δ_2	Δ_3	Δ_4	Δ_5
J_1	38	1083	28728	202176	1119744
J_2	38	1083	28728	202176	1119744
J_3	38	288	8208	202176	1119744
J_4	38	288	8208	46656	1119744
J_5	8	288	1728	46656	1119744,

all positive, so Sylvester's criterion gives $J_i \succ 0$; hence q is real stable, and homogeneity of degree 5 is immediate. Exact expansion of the determinant in the monomial basis gives

$$\begin{aligned} q = {} & 1728 m_{(5)} + 19440 m_{(4,1)} + 66555 m_{(3,2)} + 140910 m_{(3,1,1)} \\ & + 250440 m_{(2,2,1)} + 547405 m_{(2,1,1,1)} + 1182860 m_{(1^5)}. \end{aligned}$$

Inverting the Kostka matrix over $\mathbb{Z}$ in the partition order $(5),(4,1),(3,2),(3,1,1),(2,2,1),(2,1,1,1),(1^5)$ translates q into the Schur basis, yielding

$$q = 1728s_{(5)} + 17712s_{(4,1)} + 47115s_{(3,2)} + 56643s_{(3,1,1)} + 62415s_{(2,2,1)} + 56437s_{(2,1,1,1)} - 2972s_{(1^5)}.$$

The last coefficient alone also follows from the single integer identity

$$[s_{(1^5)}]q = 1728 - 2(19440) - 2(66555) + 3(140910) + 3(250440) - 4(547405) + 1182860 = -2972.$$

The accompanying source package recomputes the Schur expansion in a second, independent way, and every step uses rational arithmetic only. □

Remark 2.3 (Origin of the pencil). The matrices are not a numerical accident. Let $V = \ker B \cong S^{(3,2)}$, where B maps the permutation module on 2-subsets of $[5]$ to $\mathbb{R}^5$ by $e_{ab} \mapsto e_a + e_b$, let P_i be the rank-three projection attached to the point stabilizer of i, and let G be the Gram matrix of the chosen basis of V. Then $J_i = 2G(I + 5P_i)$, and the whole one-parameter family $\det(\sum_i x_i(I + tP_i))$ has a closed Schur expansion whose $s_{(1^5)}$ coefficient turns negative for t large; $t = 5$ clears denominators. *Appendix A develops this construction in full.*

3. Problem 37: Schur-style inequality for stable polynomials

Theorem 3.2: Posed by Borcea and Brändén (2009); counterexample found by GPT-5.6 (Pro) (2026-07-30) and bugfixed by Opus 5 (2026-08-10); formalized by S. Sra; audited by S. Sra.

Next, we close off the lower endpoint analog of Problem 1.2 (which concerned $a_\lambda \le f^\lambda a_{(1^d)}$).

Problem 3.1 (As posed: Borcea–Brändén, AIM problem list (2009), Problem 37)**.** If A is a $d \times d$ matrix then $\det(A) = \mathrm{Imm}_{(1^d)}(A)$ and $\mathrm{per}(A) = \mathrm{Imm}_{(d)}(A)$, where $(1^d) = (1,\dots,1)$ and $(d) = (1^d)' = (d,0,\dots,0)$. Schur's inequality [J,L] asserts that if A is a $d \times d$ positive semi-definite matrix then $\mathrm{Imm}_\lambda(A) \ge f^\lambda \det(A)$ for all $\lambda \vdash d$, where f^λ is the number of standard Young tableaux of shape λ. A natural real stable extension of this result would be as follows.
Problem 37. Let p be as in Problem 36. Is $a_\lambda \ge f^\lambda a_{(d)}$ for all $\lambda \vdash d$?

The lower endpoint needs a different member of the same family, because it needs a witness that is Schur *positive*. Let G be the Gram matrix of the basis of V that produced the J_i,

$$G = \begin{pmatrix} 4 & 2 & 2 & 1 & 3 \\ 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 4 & 2 & 3 \\ 1 & 2 & 2 & 4 & 3 \\ 3 & 3 & 3 & 3 & 6 \end{pmatrix},$$

put $\widetilde{J}_i := (4J_i + 2G)/5$, again a symmetric positive-definite *integer* matrix, and define

$$\widetilde{q}(x_1,\dots,x_5) = \frac{1}{5184}\det\Big(\sum_{i=1}^5 x_i \widetilde{J}_i\Big), \tag{3.1}$$

which is p_4 in the notation of Appendix A.

Theorem 3.2 (**[Disproof]**: lower endpoint, AIM Problem 37)**.** *The polynomial* (3.1) *satisfies every hypothesis of Problem 3.1, yet it violates the lower endpoint inequality $a_\lambda \ge f^\lambda a_{(d)}$: at $\lambda = (1^5)$, where $f^{(1^5)} = 1$,*

$$a_{(1^5)} = 69 < 125 = a_{(5)}.$$

Proof. The leading principal minors of the $\widetilde{J}_i$ are

	Δ_1	Δ_2	Δ_3	Δ_4	Δ_5
$\widetilde{J}_1$	32	768	17280	118800	648000
$\widetilde{J}_2$	32	768	17280	118800	648000
$\widetilde{J}_3$	32	240	5760	118800	648000
$\widetilde{J}_4$	32	240	5760	32400	648000
$\widetilde{J}_5$	8	240	1440	32400	648000,

all positive, so $\widetilde{J}_i \succ 0$ and $\widetilde{q}$ is real stable, symmetric and homogeneous of degree five by the argument used for (2.1). It can be verified that $\widetilde{q}$ has the Schur positive representation

$$\widetilde{q} = 125s_{(5)} + 1100s_{(4,1)} + 2545s_{(3,2)} + 3110s_{(3,1,1)} + 3321s_{(2,2,1)} + 2972s_{(2,1,1,1)} + 69s_{(1^5)}, \tag{3.2}$$

which can be obtained by starting with the direct monomial basis expansion

$$\widetilde{q} = 125m_{(5)} + 1225m_{(4,1)} + 3770m_{(3,2)} + 7980m_{(3,1,1)} + 13846m_{(2,2,1)} + 30004m_{(2,1,1,1)} + 64472m_{(1^5)},$$

from which upon inverting the Kostka matrix in the same partition order yields (3.2). Now $f^{(1^5)} = 1$, so the lower endpoint at $\lambda = (1^5)$ would require $69 \geq 125$. It is the only partition at which the inequality fails: the remaining six hold, the tightest being $\lambda = (4,1)$ with $1100 \geq 4 \cdot 125$. □

Appendix A. Specht modules and Borcea–Brändén's Problems 36, 37

The integer matrices of Theorems 2.2 and 3.2 are not a numerical accident, and this appendix says where they come from. They are the $t = 5$ and $t = 4$ members of one family of pencils built equivariantly on the Specht module $S^{(3,2)}$, whose $s_{(1^5)}$ coefficient is a fixed quintic in t that turns negative once t is large enough. Both problems are refuted by that single coefficient, but at different values of t, and the appendix ends by locating the two thresholds.

The module and its projections. Let M be the real permutation module with orthonormal basis $\{e_{ab} : 1 \leq a < b \leq 5\}$ indexed by the 2-subsets of $[5]$, and let

$$B : M \longrightarrow \mathbb{R}^5, \qquad B(e_{ab}) = e_a + e_b$$

be the $\mathfrak{S}_5$-equivariant map that forgets which pair produced which points. Row i of B has a 1 in the four columns ab with $i \in \{a, b\}$, and two rows share exactly one such column, so

$$BB^{\mathsf{T}} = 3I_5 + \mathbf{1}\mathbf{1}^{\mathsf{T}},$$

which is nonsingular. Hence B is surjective and $V := \ker B$ has dimension $10 - 5 = 5$. Young's rule for the permutation module on 2-subsets gives $M \cong S^{(5)} \oplus S^{(4,1)} \oplus S^{(3,2)}$, while the point-permutation module $\mathbb{R}^5$ is $S^{(5)} \oplus S^{(4,1)}$, so

$$V \cong S^{(3,2)}.$$

Write ρ for the representation of $\mathfrak{S}_5$ on V, orthogonal for the inner product V inherits from M. For $i \in [5]$ set

$$P_i := \frac{1}{2} \sum_{\substack{1 \leq a < b \leq 5 \\ a, b \neq i}} \rho((ab)), \tag{A.1}$$

half the sum of the transpositions fixing i.

Lemma A.1. *Each P_i is an orthogonal projection of rank three, $\rho(\sigma) P_i \rho(\sigma)^{-1} = P_{\sigma(i)}$ for every $\sigma \in \mathfrak{S}_5$, and $\sum_{i=1}^5 P_i = 3I_V$.*

Proof. The transpositions in (A.1) are exactly those of the point stabilizer $H_i \cong \mathfrak{S}_4$, and their sum is the class sum of transpositions, which is central in $\mathbb{R}[\mathfrak{S}_4]$ and therefore acts on an irreducible Specht module S^μ by the sum of the contents of μ. By branching,

$$S^{(3,2)}{\downarrow_{\mathfrak{S}_4}} \cong S^{(3,1)} \oplus S^{(2,2)},$$

and the content sums are $0+1+2-1=2$ for $(3,1)$ and $0+1-1+0=0$ for $(2,2)$. So the operator in (A.1) is the identity on the three-dimensional $S^{(3,1)}$ summand and zero on the two-dimensional $S^{(2,2)}$ summand: an orthogonal projection of rank three. Covariance follows by conjugating the class sum, since conjugation by $\rho(\sigma)$ carries the transpositions fixing i to those fixing $\sigma(i)$. Finally $\sum_i P_i$ commutes with $\rho(\mathfrak{S}_5)$, so Schur's lemma makes it a scalar on the irreducible V; its trace is $5 \cdot 3 = 15$, hence it is $3I_V$. □

The one-parameter family. For $t \geq 0$ put

$$A_i(t) := I_V + tP_i, \qquad p_t(x) := \det\Big(\sum\nolimits_{i=1}^{5} x_i A_i(t)\Big),$$

the determinant taken on V. By Lemma A.1 each $A_i(t)$ is symmetric positive definite with eigenvalues $1+t$ of multiplicity three and 1 of multiplicity two.

Proposition A.2. *For every $t \geq 0$ the polynomial p_t is symmetric, homogeneous of degree five, real stable, and has nonnegative coefficients in the monomial basis. Its Schur coefficients are*

$$\begin{aligned}
[s_{(5)}]p_t &= (t+1)^3,\\
[s_{(4,1)}]p_t &= \tfrac{1}{2}(t+1)^2(3t^2+8t+8),\\
[s_{(3,2)}]p_t &= \tfrac{1}{16}(t+1)(9t^4+48t^3+128t^2+160t+80),\\
[s_{(3,1,1)}]p_t &= \tfrac{1}{16}(t+1)(9t^4+64t^3+168t^2+192t+96),\\
[s_{(2,2,1)}]p_t &= \tfrac{1}{8}(6t^5+41t^4+108t^3+156t^2+120t+40),\\
[s_{(2,1,1,1)}]p_t &= \tfrac{1}{8}(6t^5+35t^4+96t^3+132t^2+96t+32),\\
[s_{(1^5)}]p_t &= -\tfrac{1}{16}(9t^5-21t^4-56t^3-72t^2-48t-16).
\end{aligned}$$

In particular $[s_{(1^5)}]p_5 = -743/2 < 0$.

Proof. Symmetry is the covariance of Lemma A.1: relabeling the variables by σ conjugates the pencil by $\rho(\sigma)$, which leaves its determinant alone. For real stability, suppose $z_1, \dots, z_5$ lie in the open upper half-plane and $\big(\sum_i z_i A_i(t)\big)v = 0$ for some $v \neq 0$. Then

$$0 = \operatorname{Im}\Big\langle v, \Big(\sum\nolimits_i z_i A_i(t)\Big)v\Big\rangle = \sum\nolimits_i \operatorname{Im}(z_i)\,\langle v, A_i(t)v\rangle > 0,$$

because each $A_i(t)$ is positive definite — a contradiction. Homogeneity of degree five is immediate. The coefficient of $x_1^{\alpha_1}\cdots x_5^{\alpha_5}$ is a positive multiple of the mixed discriminant of the multiset holding α_i copies of $A_i(t)$, and mixed discriminants of positive semidefinite matrices are nonnegative, so the monomial coefficients are nonnegative — here in fact strictly positive.

For the coefficients themselves, expanding the determinant exactly gives $p_t = \sum_{\mu\vdash 5} c_\mu(t) m_\mu$ with

$$\begin{aligned}
c_{(5)}(t) &= (t+1)^3,\\
c_{(4,1)}(t) &= \tfrac{1}{2}(t+1)^2(3t^2+10t+10),\\
c_{(3,2)}(t) &= \tfrac{1}{16}(t+1)(9t^4+72t^3+232t^2+320t+160),\\
c_{(3,1,1)}(t) &= \tfrac{1}{8}(t+1)(t+2)(9t^3+62t^2+120t+80),\\
c_{(2,2,1)}(t) &= \tfrac{1}{16}(39t^5+317t^4+1040t^3+1728t^2+1440t+480),\\
c_{(2,1,1,1)}(t) &= \tfrac{1}{8}(45t^5+348t^4+1100t^3+1764t^2+1440t+480),\\
c_{(1^5)}(t) &= \tfrac{1}{8}(99t^5+765t^4+2320t^3+3600t^2+2880t+960),
\end{aligned}$$

every numerator having positive coefficients, which is the promised positivity again. In the partition order $(5),(4,1),(3,2),(3,1,1),(2,2,1),(2,1,1,1),(1^5)$ the Kostka matrix of $s_\lambda = \sum_\mu K_{\lambda\mu} m_\mu$ is

$$K = \begin{pmatrix}
1 & 1 & 1 & 1 & 1 & 1 & 1\\
0 & 1 & 1 & 2 & 2 & 3 & 4\\
0 & 0 & 1 & 1 & 2 & 3 & 5\\
0 & 0 & 0 & 1 & 1 & 3 & 6\\
0 & 0 & 0 & 0 & 1 & 2 & 5\\
0 & 0 & 0 & 0 & 0 & 1 & 4\\
0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix},$$

and if c and a are the row vectors of monomial-symmetric and Schur coefficients then $c = aK$. Applying K^{-1} to the $c_\mu(t)$ gives the table above. The last line needs only the final column of K^{-1}, $(1,-2,-2,3,3,-4,1)^{\mathsf{T}}$:

$$\begin{aligned}
[s_{(1^5)}]p_t &= c_{(5)} - 2c_{(4,1)} - 2c_{(3,2)} + 3c_{(3,1,1)} + 3c_{(2,2,1)} - 4c_{(2,1,1,1)} + c_{(1^5)}\\
&= -\tfrac{1}{16}(9t^5 - 21t^4 - 56t^3 - 72t^2 - 48t - 16).
\end{aligned}$$

At $t=5$ this is $-(28125-13125-7000-1800-240-16)/16 = -5944/16 = -743/2$. □

Remark A.3 (Two thresholds). The family crosses the two problems at different places, and the gap between them is where Theorem 3.2 lives. Schur positivity survives exactly as long as $[s_{(1^5)}]p_t \geq 0$, that is until the single positive real zero $t_2 \approx 4.2907656153$ of $9t^5 - 21t^4 - 56t^3 - 72t^2 - 48t - 16$; past t_2 the family refutes Problem 2.1. The lower endpoint of Problem 3.1 goes earlier. Since $f^{(1^5)} = 1$ and $[s_{(5)}]p_t = (t+1)^3$, its margin at $\lambda = (1^5)$ is

$$(t+1)^3 - [s_{(1^5)}]p_t = \frac{t^2(9t^3 - 21t^2 - 40t - 24)}{16},$$

so the endpoint fails exactly past the cubic's single positive zero $t_1 \approx 3.7205453558$. On (t_1, t_2) the family is therefore still Schur positive and already short of the endpoint — the substantive refutation of Problem 3.1 — and $t=4$ is the only integer there, which is why Theorem 3.2 is stated at it. Beyond t_2 the two failures merge and stop being distinguishable, which is the reason q at $t=5$ cannot serve as a witness for the endpoint. Both decimals are stated for orientation only: no step of the proof, and no assertion in the certificate, uses a floating-point value.

Back to the integer pencil. What remains is bookkeeping. Choose a rational basis of $V = \ker B$, assemble it as the columns of a matrix U, and let $G = U^{\mathsf{T}}U$ be its Gram matrix. In that basis P_i is no longer symmetric but G-self-adjoint, $P_i^{\mathsf{T}} G = G P_i$, so with

$$J_i(t) := 2G(I + tP_i)$$

we get $J_i(t)^{\mathsf{T}} = 2(I+tP_i)^{\mathsf{T}}G = 2G(I+tP_i) = J_i(t)$: an ordinary symmetric matrix, positive definite because $A_i(t)$ is, and at $t=4$ and $t=5$ an integer one in the basis the certificate picks. Since $\det(G) = 162$ there,

$$\det\Big(\sum\nolimits_i x_i J_i(t)\Big) = 2^5 \det(G)\, p_t(x) = 5184\, p_t(x).$$

At $t=5$, writing $J_i := J_i(5)$ and dividing by 648 leaves

$$q(x) = \frac{1}{648}\det\Big(\sum\nolimits_i x_i J_i\Big) = 8\, p_5(x),$$

which is the normalization in (2.1), and $[s_{(1^5)}]q = 8\cdot(-743/2) = -2972$, the coefficient of Theorem 2.2. The factor 8 is what clears the denominators of Proposition A.2 and leaves that whole Schur expansion integral. At $t=4$ no such factor is needed — p_4 is already integral — so $\widetilde{J}_i := J_i(4)$ and

$$\widetilde{q}(x) = \frac{1}{5184}\det\Big(\sum\nolimits_i x_i \widetilde{J}_i\Big) = p_4(x),$$

the witness of Theorem 3.2, with $[s_{(1^5)}]\widetilde{q} = 69$ against $[s_{(5)}]\widetilde{q} = (4+1)^3 = 125$. The two pencils are related by $\widetilde{J}_i = (4J_i + 2G)/5$, which is how (3.1) can be written down from the J_i and G alone.

The certificate `counterexamples/aim-problems/verify_pencil.py` recomputes all of this in exact rational arithmetic: it builds V and the P_i and checks idempotence, G-self-adjointness, rank, covariance and $\sum_i P_i = 3I$; forms both pencils and certifies positive definiteness by exact leading principal minors; expands each determinant into all 126 monomials; derives K by enumerating semistandard Young tableaux rather than quoting it, and reads f^λ off its last column rather than quoting that either; inverts to get the Schur expansions and re-extracts every coefficient independently by the bialternant formula; verifies the closed formulas of Proposition A.2 as identities in t; and checks that $\widetilde{q}$ is Schur positive, that its lower endpoint fails at (1^5) and nowhere else, and that $t=4$ lies strictly between the two thresholds above.

Code and certificate

The case is the directory `counterexamples/aim-problems/` of the archive, at commit `73b416e`. Its `verify.py` recomputes every refutation in this note using exact arithmetic or rigorous interval enclosures, never floating-point evidence alone; it raises an error if any check fails and otherwise writes the certificate below. To rerun it:

```
git clone https://github.com/suvrit/count-ex-machina
cd count-ex-machina/counterexamples/aim-problems
python verify.py
```

- Theorem 1.4 (result `aim-problem-35`): exact arithmetic.
- Theorem 1.3 (result `aim-problem-38`): exact arithmetic.
- Theorem 2.2 (result `aim-problem-36`): exact arithmetic.
- Theorem 3.2 (result `aim-problem-37`): exact arithmetic.
- `artifacts/certificate-35-38.json`: Exact monomial and Schur coefficients of the stable degree-five polynomial, with the four POT-violating partitions.
- `artifacts/certificate-36-37.json`: Exact certificate of the det-pencil construction: basis, Gram data, the five integer PD matrices, and the Schur expansion with its negative coefficient.

AI usage

The credit line under the title records, for each statement, who posed it, who found its counterexample and when, and who formalized and audited it; the counterexamples here were found by GPT-5 (Pro) (2026-01-10); GPT-5.6 (Pro) (2026-07-30); GPT-5.6 (Pro) (2026-07-30) and bugfixed

by Opus 5 (2026-08-10). This note was assembled mechanically from the case's source in the archive by `tools/notes.py`, written with Claude Code; its mathematics is exactly that of the archive at the commit cited above.

Acknowledgements

The author (SS) thanks the Alexander von Humboldt (AvH) Foundation for their generous support via a Humboldt Professorship in AI.

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